

HARSH KUMAR

Technical Lead · Backend Systems · Distributed Architecture

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SUMMARY

Technical Lead and backend systems engineer with 3.5+ years of experience designing, scaling, and owning distributed financial platforms processing millions of operations daily. Sole architect of a multi-bank routing platform and first point of contact for lending operations across 3 international markets. Core strengths in asynchronous processing architectures, job orchestration, high-throughput pipelines, and cross-market platform ownership. Track record of delivering 3–5x performance improvements through system redesign, workload partitioning, and database optimization.

CORE COMPETENCIES

Distributed Systems Design	Async & Event-Driven Architecture	Job Orchestration & Pipelines
Database Partitioning & Optimization	Idempotent Processing & Recovery	Performance Engineering
Multi-Tenant Routing & API Design	Cross-Market Platform Ownership	Technical Leadership & Mentorship

KEY PROJECT

Multi-Bank Routing Interceptor | *Sole Architect & Technical Owner*

Node.js, Express.js, PostgreSQL, Redis, RabbitMQ | LMS: Ruby on Rails | Admin UI: React

- Designed and architected a stateless HTTP proxy layer that routes API and USSD traffic from multiple MNO-country entry points to isolated bank LMS instances, enabling multi-bank lending operations without modifying the existing Rails LMS.
- Engineered a smooth weighted round-robin distribution algorithm using a 100-entry slot table with atomic Redis INCR, achieving exact ratio adherence — 60 customers at 34/33/33 split precisely 20/20/20, versus 34/26/0 with naive range-based approaches.
- Solved cross-worker determinism under PM2 cluster mode by ensuring byte-identical slot tables across processes through sorted inputs, integer-only arithmetic, and fixed tie-breaking.
- Built a bulk import pipeline processing 1M+ customer records: CSV stream chunked at 1,000 rows per RabbitMQ message, consumed by 4 parallel workers doing 500-row batched upserts with Redis pipelining — 200–500x faster than sequential.
- Designed graceful degradation: unusable routing config falls back to default bank rather than refusing requests, preserving customer onboarding during misconfigurations or Redis outages.
- Implemented monthly RANGE-partitioned audit tables with auto-provisioning and PostgreSQL triggers to capture all customer reassignments including direct database changes.
- Delivered 2,157 backend tests across 65 suites with 95% coverage gate in CI; distribution logic at 100% branch coverage.

EXPERIENCE

Technical Lead

Yabx (Comviva) · Jul 2026 – Present

- First point of contact and technical owner for the core lending LMS (Ruby on Rails) across 3 international markets — 2 micro-loan markets and 1 overdraft loan market — owning end-to-end system reliability, feature delivery, and cross-market coordination.
- Architected and delivered the Multi-Bank Routing Interceptor (see Key Project), enabling the platform to support multiple banking partners per MNO-country without changes to the existing LMS.
- Drive technical decisions across backend architecture, database design, and system integration, balancing feature velocity with platform stability across markets with distinct lending products and regulatory requirements.

Senior Software Engineer

Yabx (Comviva) · Dec 2024 – Jun 2026

- Designed and scaled a distributed job orchestration platform processing 12M+ jobs/day, improving throughput by 40% through partitioned PostgreSQL tables and parallel asynchronous workers.
- Engineered financial reconciliation pipelines processing 300K+ transactions/day with idempotent workflows, automated mismatch detection, and self-healing failure recovery.
- Refactored credit rule processing engine, reducing execution time from 9 hours to 2–3 hours through parallel processing and workload partitioning across distributed workers.
- Designed consent management platform serving 2M+ users with current-state and audit-history tables for regulatory compliance.
- Mentored junior and mid-level engineers on system design patterns, code quality, and distributed systems best practices.

Software Engineer

Yabx (Comviva) · Aug 2023 – Dec 2024

- Designed asynchronous allocation service reducing product offer generation from 2 minutes to 10 seconds — a 12x improvement — by decoupling computation from the request cycle.
- Improved backend API latency by 40% through SQL query optimization, index tuning, and connection pooling on high-traffic endpoints.
- Built RESTful APIs for a loan collection portal enabling payment management, repayment schedules, and loan status workflows across multiple financial products.

Product Development Engineering Intern

Yabx (Comviva) · Feb 2023 – Aug 2023

- Resolved 18,000+ static analysis issues across backend services, improving codebase maintainability and deployment confidence.

TECHNICAL SKILLS

Languages: Ruby, JavaScript (Node.js), Python

Frameworks: Ruby on Rails, Express.js, React

Databases & Storage: PostgreSQL (partitioning, indexing, query optimization), MongoDB, Redis (caching, pub/sub, pipelining)

Messaging & Async: RabbitMQ, worker-based parallel execution, background job processing

Architecture & Patterns: Distributed systems, event-driven architecture, idempotent processing, multi-tenant routing, API gateway patterns, graceful degradation

AWARDS & RECOGNITION

- Certificate of Excellence for outstanding performance — Q2 FY 2024–25
- Team Excellence Award (3×) for exceptional collaboration and delivery — FY 2023–24 to FY 2025–26

EDUCATION

Master of Computer Applications · CGPA: 8.60

National Institute of Technology, Raipur · 2020 – 2023

Bachelor of Computer Applications · 72.66%

Maharishi Arvind Inst. of Sci. & Mgmt., Jaipur · 2016 – 2019